Econometrics

Probability: 🎲, 🔮, 🪙

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2025-10-03

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Probability

This set of slides covers concepts from Stock and Watson chapter 2.

Probabilities

Probability is a function that assigns a value in $[0, 1]$ to a *set* (representing *events*).

Consider a fair 6-sided dice.

- Outcome space: $\Omega = \{1, 2, 3, 4, 5, 6\}$
- Event: a partition of Ω.

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- Outcome space: $\Omega = \{1, 2, 3, 4, 5, 6\}$
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For example, the events A ("the outcome is even") and B ("the outcome is odd") are:

$$A = \{2, 4, 6\},$$
$$B = \{1, 3, 5\}.$$

Having a *fair* dice means that

$$\Pr(1) = \cdots = \Pr(6) = \frac{1}{6}$$

Task 1: Probabilities

Given a fair 6-sided dice , i.e. an outcome space $\Omega = \{1, 2, 3, 4, 5, 6\}$ and events

$$A = \{2, 4, 6\},$$
$$B = \{1, 3, 5\}.$$

how does one determine the ***likelihood*** (or the **probability**) of event A occuring?

Discrete Random Variables

A *Discrete Random Variable* is a function mapping outcomes to measurements.

For example, we might call

- X the number we obtain from throwing the dice once.
- X_1, X_2 the 2 numbers we obtain from throwing the dice two times.
- D an indicator (either `0`, `1`) for whether a randomly sampled person answers "yes" or "no" when asked whether they have children.

Gambler's Ruin

The dealer tosses a fair coin. If it comes up tails (`T`) you win, if it's heads (`H`)you loose.

Suppose we've seen the following sequence of tosses so far:

```
outcome:  H   H   H   H   H   H   H   H   H   H   ?
toss num: 1.  2.  3.  4.  5.  6.  7.  8.  9.  10. 11.
```

You have lost 10 times in a row now. Does this increase the probability that you will win the next round?

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```

You have lost 10 times in a row now. Does this increase the probability that you will win the next round?

Which sequence is more likely to occur?

1. `TTTTTTTTTT` (10 `T` in a row)?
2. `HTHTTHTHHT` (random occurence of `T` and `H`)

Gambler's Ruin

The probability of hitting 11 `H` in a row is

$$\Pr(11 \text{ Heads in a Row}) = \frac{1}{2^{11}} = 0.00048 = 0.05\%$$

Gambler's Ruin

The probability of hitting 11 `H` in a row is

$$\Pr(\text{11 Heads in a Row}) = \frac{1}{2^{11}} = 0.00048 = 0.05\%$$

Ok, but I already had 10 `H`! What's the next toss likely going to be given that miserable history?

$$\Pr(\text{toss 11 is Heads} | \text{10 Heads before})$$

Let's calculate that probability on the board!

Task 2: 2 Dice

Given two fair 6-sided dice 🎲🎲, what is the probability of obtaining *at least* once the face "5"?

Probability Distributions

PDF/PMF and CDF of discrete RVs

- PDF/PMF: Probability Density/Mass Function. Table listing each outcome and the associated probability of observing it.
- CDF: Cumulative Distribution Function. Probability that a given RV takes on a value *less than or equal a certain value* (requires a notion of *ordering* - e.g. what about a dice with 6 different **colors**?)

For our 6-sided fair dice

```
##   outcome pdf cdf
## 1       1 1/6 1/6
## 2       2 1/6 2/6
## 3       3 1/6 3/6
## 4       4 1/6 4/6
## 5       5 1/6 5/6
## 6       6 1/6 6/6
```

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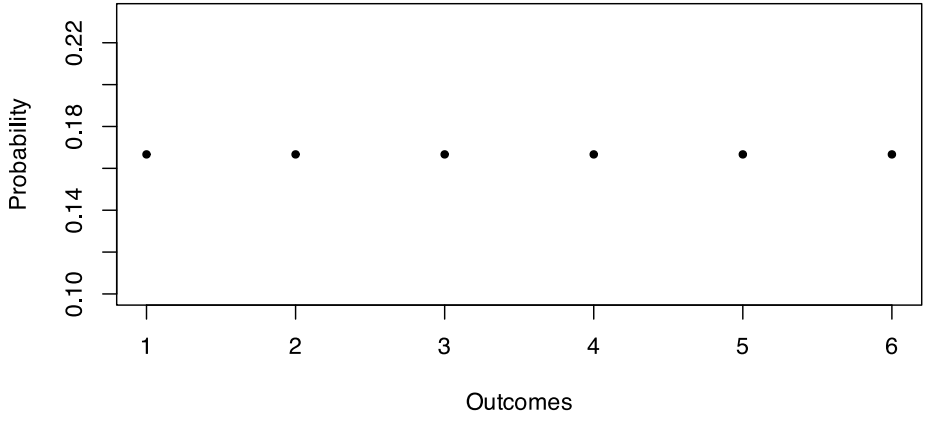

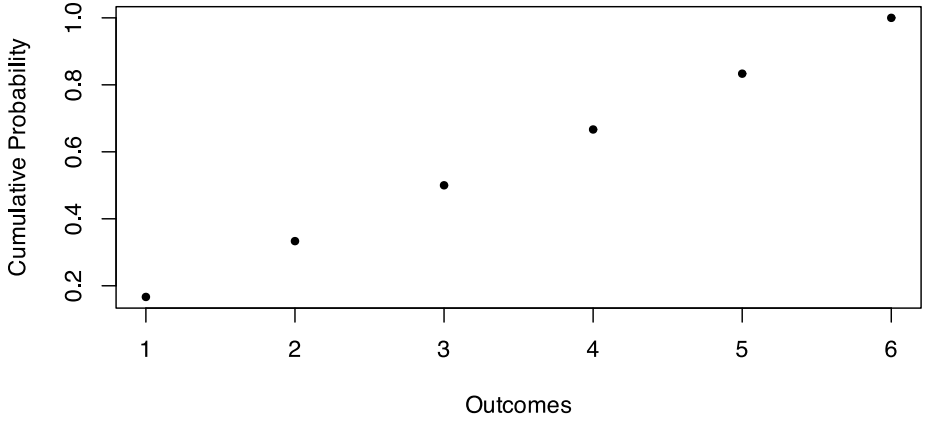

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For our fair coin

```
##   outcome pdf cdf
## 1    H(0) 1/2 1/2
## 2    T(1) 1/2 2/2
```

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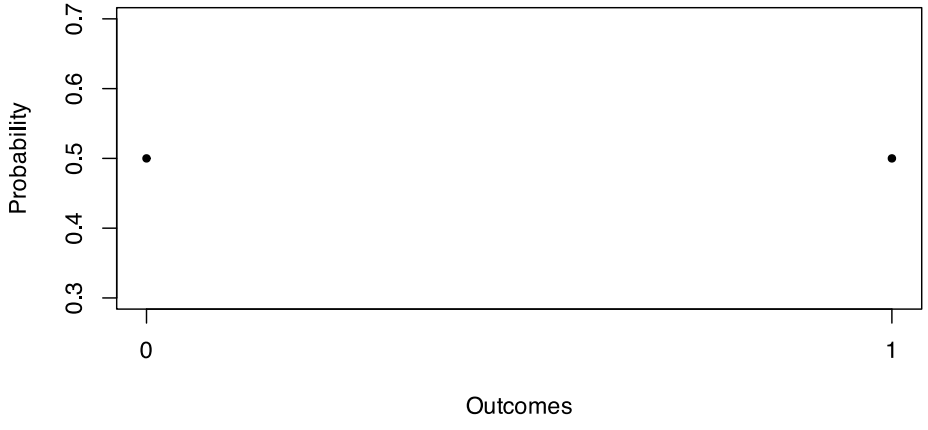
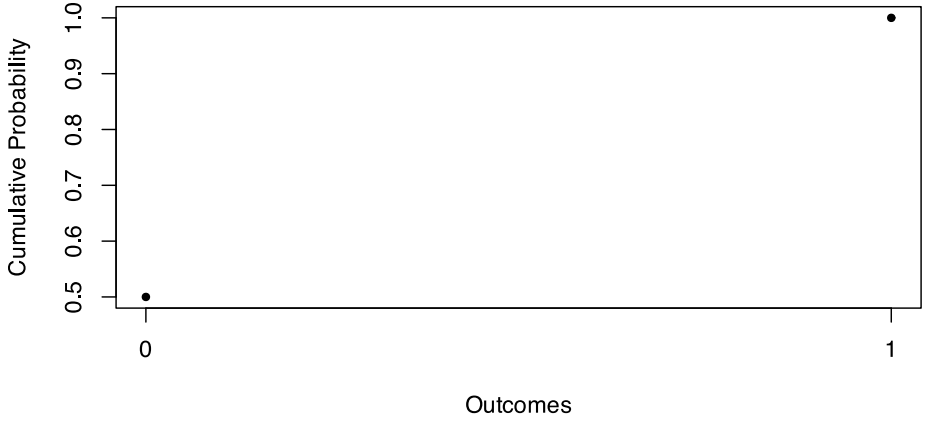
For our fair coin





Bernoulli Distribution

- A special discrete RV with 2 outcomes: `0` and `1`, where event `1` ("success") occurs with probability p.

For example:

- Tomorrow it will rain with probability p (it will *not* rain with $1 - p$).

Bernoulli Distribution

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What kind of RV is flipping a *fair* coin? What about an *unfair* coin?

PDFs and CDFs of continuous Variables

- slightly more complicated because we need calculus and integration.
- The basic idea is the same!

Expected Value and Variance

Expected Value and Variance

We write

$$E(Y) = y_1 p_1 + \cdots + y_n p_n$$

For example for our dice example, where X is the result of throwing the dice:

$$\begin{aligned} E(X) &= 1 \times \Pr(1) + 2 \times \Pr(2) + \cdots + 6 \times \Pr(6) \\ &= \end{aligned}$$

Everybody calculate this result now!

Expected Value and Variance

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For example for our dice example, where X is the result of throwing the dice:

$$\begin{aligned} E(X) &= 1 \times \Pr(1) + 2 \times \Pr(2) + \cdots + 6 \times \Pr(6) \\ &= 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + \cdots + 6 \times \frac{1}{6} \\ &= 3.5 \end{aligned}$$

- So: We need to know ***all*** weights and values ($\Pr$ and X) in order to compute this quantity.

$E(X)$ is a Theoretical Concept. Mission Impossible?

- Imagine you get this message:

$E(X)$ is a Theoretical Concept. Mission Impossible?

- Imagine you get this message:

1. Your mission, should you choose to accept it, is to inspect this device : 🔮. Whenever you touch it, it displays a number in $\{1, \ldots, 10\}$. E.g. `4,8,4,9,1,5,10,9,1`...kind of random.
2. We must know the long-run average, i.e. E(🔮). Or something terrible will happen!
3. This message will destroy itself in 10,9,8,...💣

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- You **don't know** the theoretical distribution of all the numbers in 🔮 (the $\Pr$'s). Time is running ⏰

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- You **don't know** the theoretical distribution of all the numbers in 🔮 (the $\Pr$'s). Time is running ⏰

- What could you do? 🤔

TOM CRUISE
MISSION: IMPOSSIBLE
DEAD RECKONING

$E(X)$ is a Theoretical Concept. Mission Impossible?

- If we had a huge number of observations from 🔮 (a *sample*), we could come up with a *guess* of E(🔮), based on the data we got. Empirical, like.

```
set.seed(12345)  # to ensure reproducibility
Ps = runif(10)
Ps[c(2,3)] = 0
Ps = Ps / sum(Ps) # generate random weights
x = sample(1:10, size = 10000, replace = TRUE, prob = Ps)
xbar = mean(x)
xbar
```

```
## [1] 6.3418
```

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```

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```

- `xbar` = $\bar{x} = \frac{1}{N}\sum_i^N x_i$ is called *Sample Mean*, *Arithmetic Average*, *Sample Average*
- $\bar{x}$ is an ***estimator*** for $E(X) = \mu_X$. ($E(X) = \mu_X$ by the way.)

Central Tendency - Mean and Median

`mean(x)`: the average of all values in `x`.

$$E(X) = \mu_X = \frac{1}{N}\sum_{i=1}^{N} x_i = \bar{x}$$

The second equality 👆 is correct only if...???

```
x <- c(1,2,2,2,2,100)
mean(x)
```

```
## [1] 18.16667
```

```
mean(x) == sum(x) / length(x)
```

```
## [1] FALSE
```

Central Tendency - Mean and Median

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```
## [1] 18.16667
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mean(x) == sum(x) / length(x)
```

```
## [1] FALSE
```

`median`: the value x_j below and above which 50% of the values in `x` lie. m is the median if

$$\Pr(X \leq m) \geq 0.5 \text{ and } \Pr(X \geq m) \geq 0.5$$

The median is robust against *outliers*.

```
median(x)
```

```
## [1] 2
```

Quick Review

1. EV of a bernoulli
2. Continuous RV

Variance and Standard Deviation

- A measure of *spread* of a distribution.
- The definition of *variance* is

$$var(Y) = E[(Y - \mu_Y)^2]$$

if Y is discrete,

$$var(Y) = \sum_{i=1}^{N} (y_i - \mu_Y)^2 \times p_i.$$

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- There is an issue with scaling: we *square* deviations.
- The standard deviation scales back to units of the data:

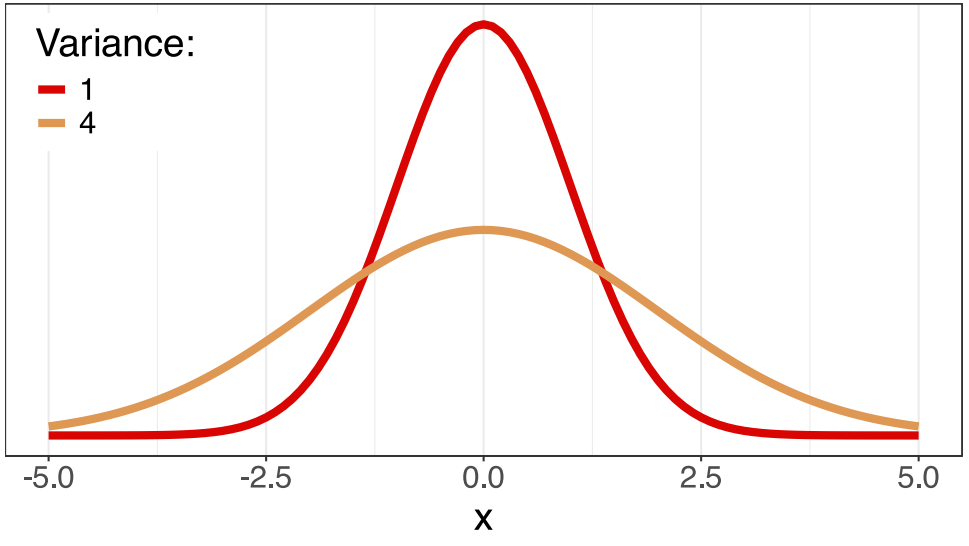
$$\sigma_Y = \sqrt{var(Y)}$$

Variance

Consider two `normal distributions` with equal mean at `0`:

Variance

Consider two `normal distributions` with equal mean at `0`:

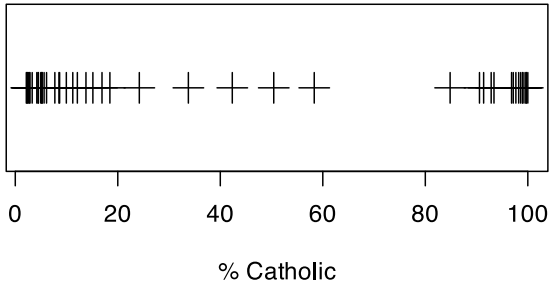


Compute with:

```
var(x)
```

Example of other Spread Measures

```
# % catholic in 47 french-speaking
# swiss cantons in 1888
plot(swiss$Catholic,rep(1,nrow(swiss)),pch = 3,
     cex = 2,xlab = "% Catholic",yaxt = "n",ylab = "")
```



How do the values in column `Catholic` *vary*?

Measure	R	Result
Variance	`var(swiss$Catholic)`	1739.29
Standard Deviation	`sd(swiss$Catholic)`	41.7
IQR	`IQR(swiss$Catholic)`	87.93
Minimum	`min(swiss$Catholic)`	2.15
Maximum	`max(swiss$Catholic)`	100
Range	`range(swiss$Catholic)`	2.15, 100

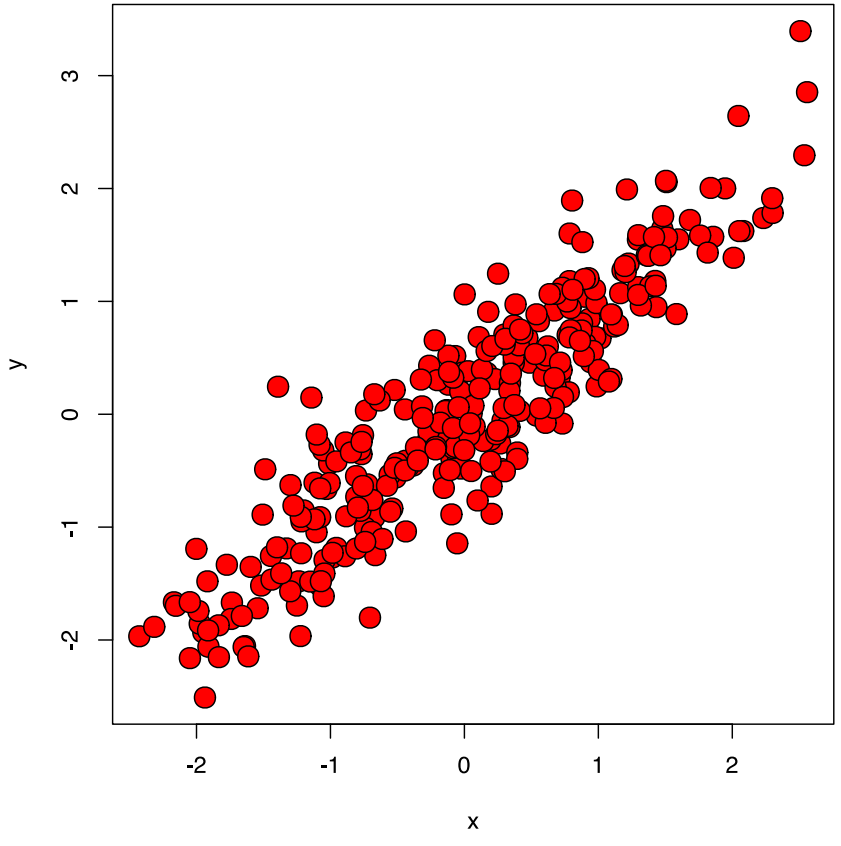
Task 3: Computing Variance by hand

1. Compute the variance of our 6-sided dice!
2. compute the variance of $X \sim \text{Bernoulli}(p)$! Bonus question: what value of p maximizes this variance?

Two Random Variables

(X, Y), (,), (,), (,)

Two Random Variables: Example



- Here, X and Y are ***joint normally*** distributed.
- We would write

$$(X, Y) \sim \mathcal{N}\Big([\, \mu_X \; \mu_Y \,] \,,\, \Sigma \Big)$$

`

where Σ is a *matrix*

$$\begin{bmatrix} \sigma_X^2 & \rho\, \sigma_X \sigma_Y \\ \rho\, \sigma_X \sigma_Y & \sigma_Y^2 \end{bmatrix}$$

Tabulating Data

`table(x)` is a useful function that counts the occurence of each unique value in `x`:

```
# install.packages("dslabs") in order to use this command
data("gapminder",package = "dslabs")
table(gapminder$continent)
```

```
##
##   Africa Americas     Asia   Europe  Oceania
##     2907     2052     2679     2223      684
```

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##
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```

The same can be achieved using the `count` function (from `dplyr`)

```
gapminder %>% count(continent)
```

```
##   continent    n
## 1    Africa 2907
## 2  Americas 2052
## 3      Asia 2679
## 4    Europe 2223
## 5   Oceania  684
```

Tabulating Data

Given two variables, `table` produces a contingency table:

```
gapminder_new <- gapminder %>%
  filter(year == 2015) %>%
  mutate(fertility_above_2 = (fertility > 2.1)) # dummy variable for fertility rate above replacement rate
```

```
table(gapminder_new$fertility_above_2)
```

```
##
## FALSE  TRUE
##    80   104
```

```
table(gapminder_new$continent,gapminder_new$fertility_above_2)
```

```
##
##            FALSE TRUE
##   Africa       2   49
##   Americas    15   20
##   Asia        20   27
##   Europe      39    0
##   Oceania      4    8
```

Cross-Tabulating Data: A joint distribution!

The probability that $X = x$ and $Y = y$ is

$$\Pr(X = x, Y = y)$$

```
# table with absolute numbers
abstab = table(gapminder_new$continent,
               gapminder_new$fertility_above_2)
(jd <- round(prop.table(abstab),2))  # proportions in
```

```
##
##            FALSE TRUE
##   Africa    0.01 0.27
##   Americas  0.08 0.11
##   Asia      0.11 0.15
##   Europe    0.21 0.00
##   Oceania   0.02 0.04
```

- check: proper probability distribution?

```
sum(jd)  # sums up all elements
```

```
## [1] 1
```

Cross-Tabulating Data: A joint distribution!

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```

- check: proper probability distribution?

```
sum(jd)  # sums up all elements
```

```
## [1] 1
```

- for instance

$$\Pr(\text{Asia}, \text{TRUE}) = 0.14673$$

- You can see 👀 that this is just the **share** of cases in each bin!

Tabulating Data: marginal distributions!

```r
# proportions by row
(m1 = prop.table(abstab, margin = 1))
```

```
##
##                 FALSE       TRUE
##   Africa   0.03921569 0.96078431
##   Americas 0.42857143 0.57142857
##   Asia     0.42553191 0.57446809
##   Europe   1.00000000 0.00000000
##   Oceania  0.33333333 0.66666667
```

check?

```r
rowSums(m1)
```

```
##   Africa Americas     Asia   Europe  Oceania
##        1        1        1        1        1
```

Tabulating Data: marginal distributions!

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(m1 = prop.table(abstab, margin = 1))
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```
##
##                 FALSE       TRUE
##   Africa   0.03921569 0.96078431
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##   Europe   1.00000000 0.00000000
##   Oceania  0.33333333 0.66666667
```

check?

```r
rowSums(m1)
```

```
##   Africa Americas     Asia   Europe  Oceania
##        1        1        1        1        1
```

```r
# proportions by column
(m2 = prop.table(abstab, margin = 2))
```

```
##
##                 FALSE       TRUE
##   Africa   0.02500000 0.47115385
##   Americas 0.18750000 0.19230769
##   Asia     0.25000000 0.25961538
##   Europe   0.48750000 0.00000000
##   Oceania  0.05000000 0.07692308
```

check?

```r
colSums(m2)
```

```
## FALSE  TRUE
##     1     1
```

Joint Distributions

The probability that $X = x$ and $Y = y$ is

$$\Pr(X = x, Y = y)$$

```
##                      Rain (X=0) No Rain (X=1)
## Long Commute (Y=0)        0.15          0.07
## Short Commute (Y=1)       0.15          0.63
```

for instance,

- $\Pr(Y = 1, X = 0) = 0.15$

Marginal Distributions

$$\Pr(Y = y) = \sum_{i=1}^{k} \Pr(X = x_i, Y = y)$$

```
##                      Rain (X=0) No Rain (X=1)
## Long Commute (Y=0)        0.15          0.07
## Short Commute (Y=1)       0.15          0.63
```

- Probability of long commute, *irrespective of rain*, is 0.15 + 0.07 = 0.22
- Probability of rain, *irrespective of length of commute*, is 0.15 + 0.15 = 0.3

Task 4 : Computing Marginal Distributions

05:00

```
r
```

```
##                      Rain (X=0) No Rain (X=1)
## Long Commute (Y=0)        0.15          0.07
## Short Commute (Y=1)       0.15          0.63
```

1. Compute the marginal distribution of X
2. Compute the marginal distribution of Y

Conditional distributions

- Conditional distribution is $\Pr(X = x | Y = y) = \frac{\Pr(X=x, Y=y)}{\Pr(Y=y)}$
- We **fix** the value of one variable, and look at the resulting distribution of the other variable.
- Notice, the result must be a *valid* probability distribution, i.e., it must sum to 1.
- We compute the probability as before, but restrict attention to where the condition we impose is true.

Task 5: compute conditional distribution

03:00

Consider

```
##                      Rain (X=0) No Rain (X=1)
## Long Commute (Y=0)        0.15          0.07
## Short Commute (Y=1)       0.15          0.63
```

- Suppose we know that it does not rain $x = 1$. What is the distribution of Y, *given* this knowledge, i.e. what is the distribution $\Pr(Y|X = 1)$?

Task 6: Computing More Conditional Distributions

05:00

```
jd
```

```
##
##             FALSE TRUE
##   Africa     0.01 0.27
##   Americas   0.08 0.11
##   Asia       0.11 0.15
##   Europe     0.21 0.00
##   Oceania    0.02 0.04
```

1. Compute the Distribution of countries conditional on low fertility?
2. Compute the Distribution of countries conditional on being in Asia?
3. Compute the marginal distribution of high/low fertility

Conditional Expectation

We define

$$E(Y|X=x) = \sum_{i=1}^{k} y_i \Pr(Y = y_i | X = x)$$

- Notice the y_i there!
- Basically this is expected value of Y, but **under the assumption that** X takes on the value x.
- We *fix* the joint distribution of (Y, X) at a certain value $X = x$.

Task 7: Conditional Expectation

TABLE 2.3 Joint and Conditional Distributions of Number of Wireless Connection Failures (M) and Network Age (A)

A. Joint Distribution

	$M = 0$	$M = 1$	$M = 2$	$M = 3$	$M = 4$	Total
Old network ($A = 0$)	0.35	0.065	0.05	0.025	0.01	0.50
New network ($A = 1$)	0.45	0.035	0.01	0.005	0.00	0.50
Total	0.80	0.10	0.06	0.03	0.01	1.00

B. Conditional Distributions of M given A

	$M = 0$	$M = 1$	$M = 2$	$M = 3$	$M = 4$	Total
$\Pr(M \mid A = 0)$	0.70	0.13	0.10	0.05	0.02	1.00
$\Pr(M \mid A = 1)$	0.90	0.07	0.02	0.01	0.00	1.00

1. Compute $E(M)$
2. Compute $E(M|A = 0)$

Law of Iterated Expectations

- We can decompose the expected value of Y into subgroups, see what fraction of total probability each group makes up, and compute their weighted average.

Law of Iterated Expectations

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```
##    name sex height
## 1 Peter   M   1.90
## 2 James   M   1.75
## 3  Mary   F   1.68
## 4  John   M   1.86
## 5   Amy   F   1.59
```

- The average height here is $(1.9 + 1.75 + 1.68 + 1.86 + 1.59)/5 = 1.756$:

Law of Iterated Expectations

> The average height here is 1.756

- What is the average for each **group** by sex?

```
## # A tibble: 2 × 3
##   sex       n mean_height
##   <chr> <int>       <dbl>
## 1 F         2        1.64
## 2 M         3        1.84
```

Law of Iterated Expectations

> The average height here is 1.756

- What is the average for each **group** by sex?

```
## # A tibble: 2 × 3
##   sex       n mean_height
##   <chr> <int>       <dbl>
## 1 F         2        1.64
## 2 M         3        1.84
```

- Can we recover the unconditional mean from this information? Yes we can!
- We need again *weights* and *values*, like for the standard $E(Y)$

```r
weights = c(2/5, 3/5)
values = c(1.635, 1.836667)
sum(weights * values) # 2/5 * 1.635 + 3/5 * 1.836667
```

```
## [1] 1.756
```

Law of Iterated Expectations

- What did we just do?
- We established that we can get the unconditional mean of Y from the *conditional* one, if we know the *weights* we have to give to each subgroup. More precisely:

1. We know $E(Y|X)$:
    1. $E(Y|\text{male}) = 1.836667$
    2. $E(Y|\text{female}) = 1.635$
2. We know the proportion of both groups:
    1. $\Pr(\text{male}) = 3/5$
    2. $\Pr(\text{female}) = 2/5$

Therefore, we can recompose $E(Y)$:

$$\begin{aligned} E(Y) &= \sum_{i=1}^{k} E(Y|X = x_i) \Pr(X = x_i) \\ &= E_X \left[E(Y|X) \right] \end{aligned}$$

Law of Iterated Expectations

- One particular version of this comes up often in econometrics:

$$E[Y|X] = 0 \Rightarrow E[Y] = 0$$

- Why? Well, now you know that $E[Y|X] = 0$ for all possible X is zero.
- Compute $E(Y) = \sum_{i=1}^{k} 0 \times \Pr(X = x_i)$!

Conditional Expectation and Prediction Errors

- We said that the the squared error makes sense (talking about variance)
- Suppose we want to design a **predictor** $g(X)$ which can **predict** Y reasonably well. One criterion for that might be the *average squared error of prediction*.

Conditional Expectation and Prediction Errors

- We said that the the squared error makes sense (talking about variance)
- Suppose we want to design a **predictor** $g(X)$ which can **predict** Y reasonably well. One criterion for that might be the *average squared error of prediction*.
- In other words, what is the solution to this problem?

$$\min_m E\left[(Y-m)^2\right]$$

Independence

- We call 2 RVs *independent* if knowing the value of one of them is not informative about the value of the other.
- The precise definition of this statement is

$$\Pr(X = x | Y = y) = \Pr(X = x)$$

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$$\Pr(X = x | Y = y) = \Pr(X = x)$$

- or:

$$\Pr(X = x, Y = y) = \Pr(X = x) \Pr(Y = y)$$

Covariance

- Covariance tells us how two RVs co-vary.

$$
\begin{aligned}
cov(X, Y) = \sigma_{XY} &= E\left[(X - \mu_X)(Y - \mu_Y)\right] \\
&= \sum_{i=1}^{k} \sum_{j=1}^{m} (x_i - \mu_X)(y_j - \mu_Y) \Pr(X = x_i, Y = y_j)
\end{aligned}
$$

- Notice how this object tells us how large deviations of X from its mean *jointly* occur with such deviations of Y.

- This confounds the scale of X and Y, however, so not easy to interpret

Correlation

- Correlation fixes this issue.

$$corr(X,Y) = \frac{cov(X,Y)}{\sqrt{var(X)var(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

- Correlation is always between -1 and 1.
- $corr(X,Y) = 0$ means they are *uncorrelated*.

Correlation and Conditional Mean

- If the conditional mean of Y does **not** depend on X, then they must be uncorrelated.
- That is, if

$$E(Y|X) = \mu_Y \Rightarrow cov(X, Y) = 0 \text{ and } corr(X, Y) = 0$$

Task 8 : Show Correlation and Conditional Mean

Show this result!

$$E(Y|X) = \mu_Y \Rightarrow cov(X, Y) = 0 \text{ and } corr(X, Y) = 0$$

Random Sampling (SW KC 2.5)

- Suppose we **randomly** select n objects from a large population.
- Y_i is the value of i-th object chosen.
- Each object i has the same chance of getting drawn, which means that the *distribution of the Ys is the same*, and object i being drawn does not depend on object j being drawn.
- **Same** means **identical**. So, the Ys are ***identically and independently*** distributed.

Sampling Distribution of Sample Average

- We already defined the **sample average**:

$$\bar{y} = \frac{1}{n}(y_1 + y_2 + \cdots + y_n) = \frac{1}{n}\sum_{i=1}^{n} y_i$$

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- A core concept is to realize that the value of $\bar{y}$ is itself ***random***. Just imagine drawing a second sample of y's
- Because $\bar{y}$ varies with each sample, the object $\bar{y}$ has a **sampling distribution**.
- Every well-defined distribution has a **mean** and a **variance**.

Mean of the Sample Average

- The mean of the sample average (*the mean of the mean!*) is just

$$E(\bar{y}) = E[\frac{1}{n}\sum_{i=1}^{n} y_i] = \frac{1}{n}\sum_{i=1}^{n} E[y_i] = \frac{1}{n} n\mu_Y$$

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- So that's good news. The expected value is equal to the population mean.
- This means that $\bar{y}$ is an ***unbiased estimator*** for $E(y)$.

Task 9: Variance of the Sample Average

- Similarly,

$$var(\bar{y}) = var\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right) = \frac{\sigma_Y^2}{n}$$

- Show this result!

Properties of the Sample Mean

The following facts **always** hold for the sample mean:

1. $E(\bar{y}) = \mu_Y$
2. $var(\bar{y}) = frac\sigma_Y^2 n$
3. $std.dev(\bar{y}) = frac\sigma_Y \sqrt{n}$

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The following facts **always** hold for the sample mean:

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Those are true ***whenever y is i.i.d***. In particular, we do **not** need the y to be normally distributed!

Sampling from a Normal Distribution

NOT Sampling from a Normal Distribution

The Law of Large Numbers

The Central Limit Theorem

10:00

```
library(CLT)
CLT()
```

class: inverse

Task 4: Summarising data

1. Compute the mean of GDP in 2011 and assign to object `mean`. You should exclude missing values. (*Hint: read the help for* `mean` *to remove* `NA`*s*).

2. Compute the median of GDP in 2011 and assign to object `median`. Again, you should exclude missing values. Is it greater or smaller than the average?

3. Create a density plot of GDP in 2011 using `geom_density`. A density plot is a way of representing the distribution of a numeric variable. Add the following code to your plot to show the median and mean as vertical lines. What do you observe?
   `geom_vline(xintercept = as.numeric(mean), colour = "red") +`
   `geom_vline(xintercept = as.numeric(median), colour = "orange")`

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That's all for Probability!

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